

# HW7 Solutions

ECE 310: Digital Signals Processing, Spring 2025

Version: 1.0

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## Solution:

The relation between the discrete signal and the continuous time signal is  $x[n] = x_c(\Omega_0 nT)$ .

For a discrete cosine signal with a digital radial frequency of  $\pi/4$ , possible frequencies can be

$$\Omega_0 = \frac{1}{T} \left( \frac{\pi}{4} + 2k\pi \right), k \in \mathbb{Z}_{\geq 0},$$

or

$$\Omega_0 = \frac{1}{T} \left( \frac{-\pi}{4} + 2k\pi \right), k \in \mathbb{Z}_{>0}.$$

For example, the two smallest possible frequencies are  $1000\pi/4$  and  $-1000\pi/4$ .

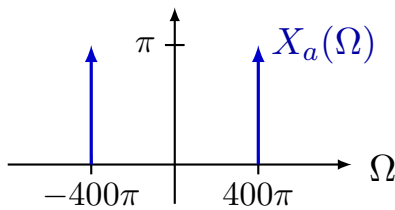
**Grading:** 12 points

- +6 for the first correct frequency
- +6 for another correct frequency

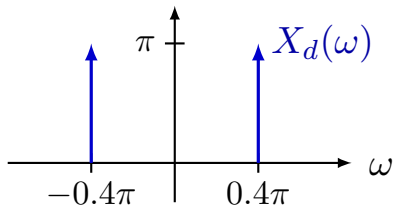
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## Solution:

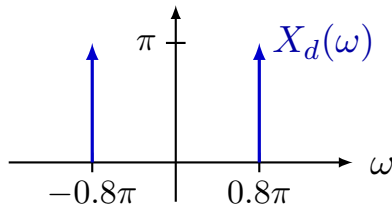
a)  $x_a(t) = \cos(400\pi t) \xleftrightarrow{\text{CTFT}} \pi[\delta(\Omega - 400\pi) + \delta(\Omega + 400\pi)].$



For the discrete case,  $x[n] = \cos(400\pi nT) = \cos(0.4\pi n) \xleftrightarrow{\text{DTFT}} \pi[\delta(\omega - 0.4\pi) + \delta(\omega + 0.4\pi)].$



- b) A larger sampling period does not change the continuous signal, so the  $x_a(\Omega)$  remains the same as the previous result. The discrete-time signal does change,  $x[n] = \cos(400\pi n * \frac{2}{1000}) = \cos(0.8\pi n) \xleftrightarrow{\text{DTFT}} \pi[\delta(\omega - 0.8\pi) + \delta(\omega + 0.8\pi)].$



- c) The maximum frequency for  $X_a$  is  $f_{\max} = 400\pi/2\pi = 200\text{ s}^{-1}$ . According to the Nyquist criteria, i.e.  $1/T > 2f_{\max}$ , the maximum sampling period  $T_{\max} = 1/400 = 2.5\text{ ms}$ .

**Grading:** 25 points

- (a)
  - +5 for correct plot of  $X_a(\Omega)$
  - +5 for correct plot of  $X_d(\omega)$
- (b)
  - +3 for correct plot of  $X_a(\Omega)$
  - +5 for correct plot of  $X_d(\omega)$
- (c)
  - +4 for correct use of Nyquist criteria
  - +3 for correct result

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#### Solution:

Recall the following characteristics when we move from the CTFT to DTFT:

1.  $\omega = \Omega T$
2. Amplitude scaling factor of  $1/T$  (Except for  $\delta$ -functions)

One more thing to keep in mind is that while CTFTs are not periodic, DTFTs are. As such, we need to be careful of aliasing and factor these into our graphs.

Using these facts, we can create our graphs:

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> **a)**  $T = 1/100$  (sec)

Recall that  $\omega = 2\pi/T$  (since angular frequency is used in the provided graph).

Thus our Nyquist Rate is  $2 \cdot \text{largest } \omega = 2 \cdot 2\pi/T = 400\pi$ . Since we do not meet this criterion, we will experience aliasing.

We first draw both the "original copy" as well as our spectral copies that reside at  $2\pi$  and  $-2\pi$  (dotted black line) and use our 2 characteristics to graph the DTFT ( $X_d(\omega)$ , solid black line):

Now we can "add" the spectral copies and just graph 1 period ( $[-\pi, \pi]$ ):

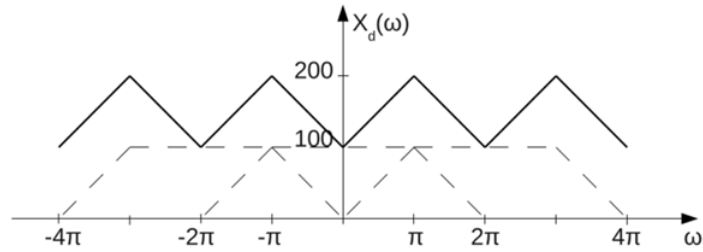


Figure 1: a) DTFT Graph

> b)  $T = 1/200$  (sec)

Now we meet the Nyquist Rate and don't have to worry about aliasing. We directly graph a single period here.

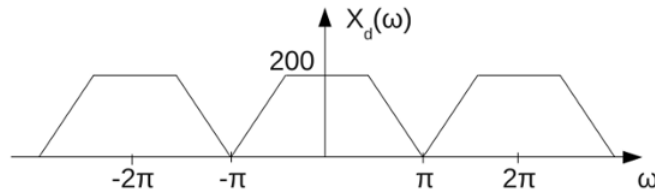


Figure 2: b) DTFT Graph

**Grading:** 25 points

- +15 (a)
  - +4 correct amplitude for base copy
  - +4 correct  $\omega_n$  for base copy
  - +5 consider aliasing
  - +2 correct final graph
- +10 (b)
  - +4 correct amplitude for base copy
  - +4 correct  $\omega_n$  for base copy
  - +2 correct final graph

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### Solution:

Consider  $x_{sampled}(t)$ , which is the result of multiplying  $x_a(t)$  with the impulse train  $\sum_{n=-\infty}^{\infty} \delta(t - nT)$ :

$$x_{sampled}(t) = x_a(t) \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Denote the FT of  $x_{sampled}(t)$  as  $X_s(\Omega)$ .

According to the windowing property  $x_a(t)w(t) \leftrightarrow \frac{1}{2\pi} X_a(\Omega) * W(\Omega)$ ,

and the transform pair  $\sum_{n=-\infty}^{\infty} \delta(t - nT) \leftrightarrow \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta(\Omega - n\frac{2\pi}{T})$ , we can obtain:

$$\begin{aligned} X_s(\Omega) &= \frac{1}{2\pi} X_a(\Omega) * \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta(\Omega - n\frac{2\pi}{T}) \\ &= \frac{1}{T} \sum_{n=-\infty}^{\infty} X_a(\Omega - n\frac{2\pi}{T}) \end{aligned}$$

Then we can obtain DTFT of  $x[n]$  by using the relationship  $\omega = \Omega T$ :

$$\begin{aligned} X_d(\omega) &= X_s(\Omega)|_{\Omega=\omega/T} \\ &= \frac{1}{T} \sum_{n=-\infty}^{\infty} X_a(\frac{\omega - 2n\pi}{T}) \\ &= \frac{1}{T} \sum_{l=-\infty}^{\infty} X_a(\frac{\omega + 2l\pi}{T}) \end{aligned}$$

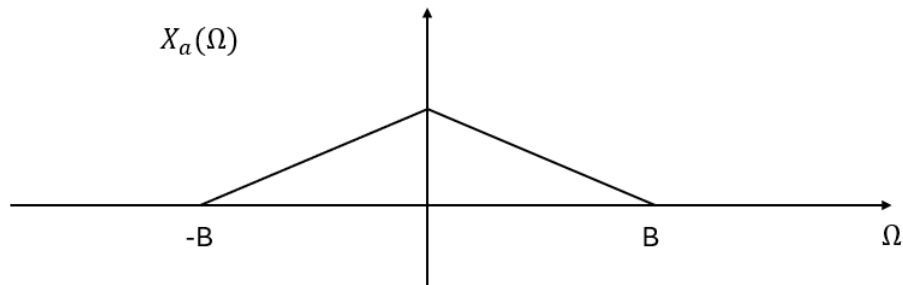
**Grading:** 23 points

- +5 getting  $x_{sampled}(t)$
- +13 getting  $X_s(\Omega)$  (+10 if minor errors)
- +5 getting  $X_d(\omega)$
- there might be other valid proofs

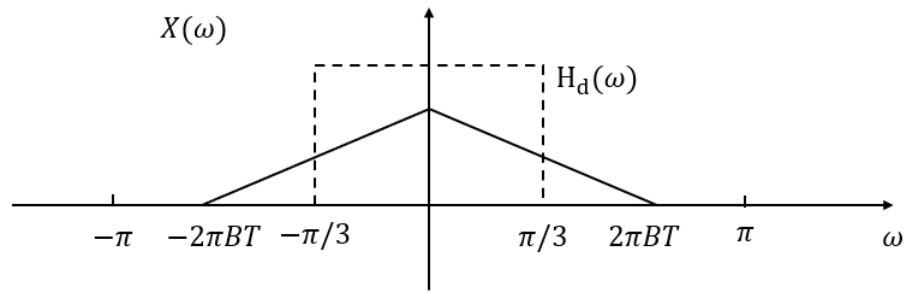
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### Solution:

We first draw bandlimited signal  $X_a(\Omega)$ , where  $B = 5$  kHz.



After A/D, we have  $X(\omega)$  as shown below.



Considering that after applying bandpass filter  $H_d(\omega)$ , the frequency above  $f_0 = 4$  kHz will be filtered out, then we should have  $2\pi f_0 T = \pi/3$ , then we have  $T = 1/(6f_0) = 1/24$  ms.

**Grading:** 15 points

- +10 getting  $2\pi f_0 T = \pi/3$
- +5 getting  $T$  value